Uniform Distribution

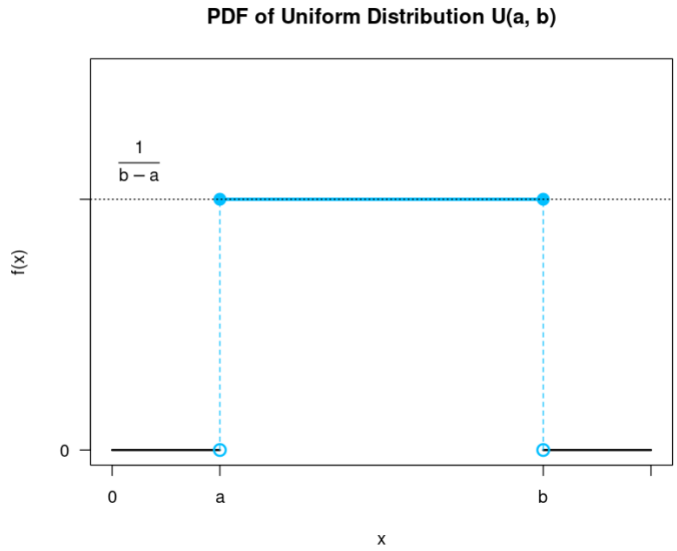
A continuous random variable X is said to have a Uniform (or Rectangular) distribution over (a, b) if its pdf is given by,

$$f(x) = \text{ constant, say } k, \quad x \in (a, b)$$
$$1 = \int_a^b f(x)dx \Rightarrow k \int_a^b dx = 1$$
$$\Rightarrow k = \frac{1}{b-a}$$
$$f(x) = \frac{1}{b-a} = \frac{1}{\text{length of interval } I}$$

1 Definition

A continuous random variable X is said to follow a Uniform distribution over (a, b) if its PDF is given by,

$$f_X(x) = \begin{cases} \frac{1}{b-a} & , a < x < b \\ 0 & , \text{ otherwise} \end{cases}$$



The graph of the pdf looks like a rectangle, that's why the distribution is known as Rectangular Distribution.

Notation

$X \sim U(a, b)$ or $X \sim \text{Rect}(a, b)$. Density $f_X(x)$ can be greater than 1.

2 CDF

$F(x) = \int_{-\infty}^{x} f(t)dt.$

$$= \begin{cases} 0 & , x < a \\ \int_a^x \frac{1}{b-a}dt & , a \leqslant x < b \\ \int_a^b \frac{1}{b-a}dt & , x \geqslant b \end{cases} = \begin{cases} 0 & , x < a \\ \frac{x-a}{b-a} & , a \leqslant x < b \\ 1 & , x \geqslant b \end{cases}$$

1. Let $X \sim U(0,1)$, does $E\left(\frac{1}{X}\right)$ exist?

$$\begin{aligned} E\left(\frac{1}{X}\right) = \int_0^1 \frac{1}{x}dx &= \lim_{a\to 0^+} \int_a^1 \frac{1}{x}dx \\ &= \lim_{a\to 0^+} [\log|x|]_a^1 \\ &= \lim_{a\to 0^+} (\ln 1 - \ln a) \\ &= \lim_{a\to 0^+} \ln \frac{1}{a} \\ &= \infty \end{aligned}$$

$\therefore E\left(\frac{1}{X}\right)$ does not exist.

2. Let $X \sim U(0,n), \quad n \in \mathbb{N}$, let $Y = X - [X]$, show that $Y \sim U(0,1)$.

$$\begin{aligned} F_Y(y) = P(Y \leqslant y) &= P[X - [x] \leqslant y] \\ &= \sum_{k=0}^{n-1} P[X \leqslant [x] + y, k \leqslant X < k+1], \quad [x] = k, \text{ say} \\ &= \sum_{k=0}^{n-1} P[X \leqslant k + y, k \leqslant X < k+1] \\ &= \sum_{k=0}^{n-1} P[k \leqslant X \leqslant k + y] \\ &= \sum_{k=0}^{n-1} \int_k^{k+y} \frac{1}{n}dx, \quad 0 < y < 1 \\ &= \sum_{k=0}^{n-1} \frac{y}{n} \\ &= y, \quad 0 < y < 1 \end{aligned}$$

$$F_Y(y) = \begin{cases} 0, y < 0 \\ 1, 0 \leqslant y < 1 \\ 1, y \geqslant 1 \end{cases} \qquad f_Y(y) = \begin{cases} 1, & 0 < y < 1 \\ 0, & ow \end{cases} \qquad \Rightarrow Y = X - [X] \sim U(0,1)$$

3. If $X \sim U\{0,1,\ldots,n\}$ and $Y \sim U(0,1)$. Find the CDF of $Z = X + Y$ and find ρ_{XZ}.

Z can take any value in interval $(0, n+1)$

$F_Z(z) = P(Z \leq z) = P(X + Y \leq z)$

$$= \begin{cases} 0 & , z \leqslant 0 \\ P(X = 0, Y \leqslant z) & , 0 \leqslant z < 1 \\ P(X = 0, Y \leqslant 1) + P(X = 1, Y \leqslant z - 1) & , 1 \leqslant z < 2 \\ P(X = 0, Y \leqslant 1) + P(X = 1, Y \leqslant 1) + P(X = 2, Y \leqslant z - 2) & , 2 \leqslant z < 3 \\ P(X = 0, Y \leqslant 1) + P(X = 1, Y \leqslant 1) + \cdots + P(X = k - 1, Y \leqslant 1) & , k \leqslant z < k + 1 \end{cases}$$

$$= \begin{cases} 0 & , z < 0 \\ \frac{k}{n+1} + \frac{1}{n+1}(z - k) & , k \leqslant z < k + 1 \\ 1 & , z \geqslant n + 1 \end{cases}$$

$$f_Z(z) = \begin{cases} \frac{1}{n+1}, & 0 \leqslant z < n + 1, \\ 0, & \text{otherwise} \end{cases}$$

$\therefore Z = X + Y \sim U(0, n + 1)$

4. Prajwel and his girlfriend decide to meet at a certain location. If each of them arrives at a time uniformly distributed between 12 noon to 1 pm. Find the probability that the first one to arrive has to wait larger than 10 minutes.

Let, X and Y be the arrival time in minutes of prajwel and his gf after 12 noon.

$$X \sim U(0, 60)$$

$$p(|X - Y| > 10) = \iint_{|X-Y|>10} f_{X,Y}(x, y)dxdy$$

$$= \iint_A \frac{1}{60^2}dxdy \quad A = \{(x, y) : |x - y| > 60; 0 \leq x, y \leq 60\}$$

$$= \frac{\iint A dxdy}{60^2}$$

$$= \frac{\text{Bounded by the region } A}{\text{The area bounded by } \Omega}$$

$$= \frac{50^2}{60^2} = \frac{5}{6}$$

$\int_a^b \int_c^d f(x, y)dxdy =$ base area $\times h$

$\Rightarrow h \int_a^b \int_c^d dxdy =$ base area $\times h$

$\Rightarrow \int_a^b \int_c^d dxdy =$ Area

3 Median

Let Me be the median of the distribution.

By definition, $\int_\alpha^{Me} f(x)dx = \int_{Me}^B f(x)dx = \frac{1}{2}$

$$\Rightarrow \int_{Me}^{\beta} f(x)dx = \frac{1}{2}$$

$$\Rightarrow \frac{\beta - Me}{\beta - \alpha} = \frac{1}{2}$$

$$\Rightarrow \beta - Me = \frac{\beta - \alpha}{2}$$

$$\Rightarrow Me = \beta - \frac{\beta - \alpha}{2} = \frac{\beta + \alpha}{2}$$

$$\text{Mean } = \text{ median } = \frac{\alpha + \beta}{2}$$

Mode does not exist for uniform distribution.

4 Quartile Deviation

Let Q_1, Q_3 be the 1st and 3rd quartile of the distribution.

By definition,

$$\begin{aligned}
\int_\alpha^{Q_1} f(x)dx = 1/4 \qquad & \int_\alpha^{Q_3} f(x)dx = \tfrac{3}{4} \\
\Rightarrow \frac{Q_1-\alpha}{\beta-\alpha} = \frac{1}{4} \qquad & \Rightarrow \frac{Q_3-\alpha}{\beta-\alpha} = \frac{3}{4} \\
\Rightarrow Q_1 = \frac{\beta-\alpha}{4} + \alpha \qquad & \Rightarrow Q_3 = \tfrac{3}{4}(\beta-\alpha)+\alpha \\
= \frac{3\alpha+\beta}{4} \qquad & = \frac{3\beta+\alpha}{4}
\end{aligned}$$

$$QD = \frac{Q_3-Q_1}{2} = \frac{3\beta+\alpha-3\alpha-\beta}{8}$$

$$= \frac{2\beta-2\alpha}{8} = \frac{\beta-\alpha}{4}$$

Remark

Consider a Uniform distribution over the region $\Omega \in \mathbb{R}^n$, then the PDF of X is,

$$f(x) = \begin{cases} c & x \in \Omega \\ 0 & x \in \mathbb{R}^n - \Omega \end{cases}$$

Then, $1 = \int_\Omega f(x)dx \Rightarrow c\int_\Omega f(x)dx = 1$

$$\Rightarrow c = \frac{1}{\text{Measure of } \Omega}$$

The probability that one randomly selected point $\underset{\sim}{x}$ falls in a region $A \subset \mathbb{R}^n$

$$\begin{aligned}
= P(\underset{\sim}{x} \in A) &= \int_A f(\underset{\sim}{x})d\underset{\sim}{x} \\
&= c\int_A d\underset{\sim}{x} \\
&= \frac{\text{The measure of } A}{\text{The measure of } \Omega}
\end{aligned}$$

5 Moments

$$\begin{aligned}
\mu_r' = E\left(X^r\right) &= \int_a^b x^r \frac{1}{b-a}dx \\
&= \frac{x^{r+1}}{(r+1)(b-a)}\Big|_a^b \\
&= \frac{b^{b+1}-a^{b+1}}{(b-a)(r+1)} \\
E(X) = \mu_1' &= \frac{b+a}{2} \\
V(X) = \mu_2' - \mu_1'^2 &= \frac{b^3-a^3}{3(b-a)} - \frac{(b+a)^2}{4} \\
&= \frac{(b-a)^2}{12}
\end{aligned}$$

5. Let X be a RV denifed on [0,1] if $P(x < X < y) \propto (y - x) \quad \forall \quad 0 < x < y < 1$, show that $X \sim U(0, 1)$.

Let $F(x)$ be the CDF of X, $\quad 0 \leqslant x \leqslant y \leqslant 1$

$$
\begin{aligned}
& P(x \leqslant X \leqslant y) = k(y - x) \\
\Rightarrow\ & F(y) - F(x) = k(y - x) \\
\Rightarrow\ & \frac{F(y) - F(x)}{y - x} = k \quad \forall x < y \\
\Rightarrow\ & \lim_{y \to x} \frac{F(y) - F(x)}{y - x} = k \\
\Rightarrow\ & F'(x) = k \\
\Rightarrow\ & F(x) = kx + c, \quad 0 \leqslant x \leqslant 1 \\
& F(0) = 0 \Rightarrow c = 0 \\
& F(1) = 1 \Rightarrow k = 1.
\end{aligned}
$$

The CDF of X is $F_X(x) = \begin{cases} x, & 0 \leqslant x < 1 \\ 1, & x \geqslant 1 \end{cases}$

The PDF of X is $f_X(x) = \begin{cases} 1 & , 0 < x < 1 \\ 0 & , \text{otherwise} \end{cases}$

$$\Rightarrow X \sim U(0, 1)$$